

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Programming Fundamentals and Data Structures	Course Code: CMPE-112L
Assignment Type: Lab Report	Dated: 14th May, 2026
Semester: 1st	Session: 2026
Lab/Project/Assignment #: 14	CLOs to be covered: CLO2
Lab Title: OS Library, if __name__ == "__main__", is vs ==	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO2	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Mark s	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 14: OS Library, if name == " main ", is vs ==

Lab Goals/Objectives:

To understand the use of the os library for file/directory operations, the purpose of the __name__ == "__main__" guard, and basic file handling operations (read, write, append).

Equipment Required:

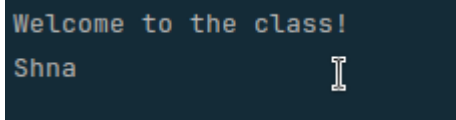
Python 3, VS Code / PyCharm

Theory:

The if `__name__ == "__main__"`: guard allows a Python file to be run directly or imported as a module without automatically executing its top-level code. The `os` library provides functions to interact with the operating system, such as creating directories with `os.mkdir()`. Python also provides built-in file handling using `open()` with modes such as "w" (write), "r" (read), and "a" (append).

Lab Work:**Task 1: Module Import and `__name__ == "__main__"`**

```
def welcome():  
    print("Welcome to the class!")  
    return  
if __name__ == "__main__":  
    welcome()  
def ushna():  
    print("Shna")  
    return  
import anayaazam  
anayaazam.welcome()  
anayaazam.ushna()
```

Result:

```
Welcome to the class!  
Shna
```

Task 2: Write to a File

```
a = open("data.txt", "w")  
s = a.write("welcome")  
a.close()  
print(s)
```

Result:

```
7  
welcome
```

Task 3: Read from a File

```
a = open("data.txt", "r")  
s = a.read()  
a.close()  
print(s)
```

Result:

A dark-themed terminal window showing the output of the code. The word "welcome" is displayed in a light blue font, followed by a white cursor icon (a vertical bar with a horizontal crossbar).

Task 4: Append to a File

```
a = open("data.txt", "a")  
s = a.write("hello")  
a.close()  
print(s)
```

Result:

A dark-themed terminal window showing the output of the code. The number "5" is displayed in a light blue font.

Task 5: Create Directory using os.mkdir

```
import os  
os.mkdir("anayaazam")
```

Result:



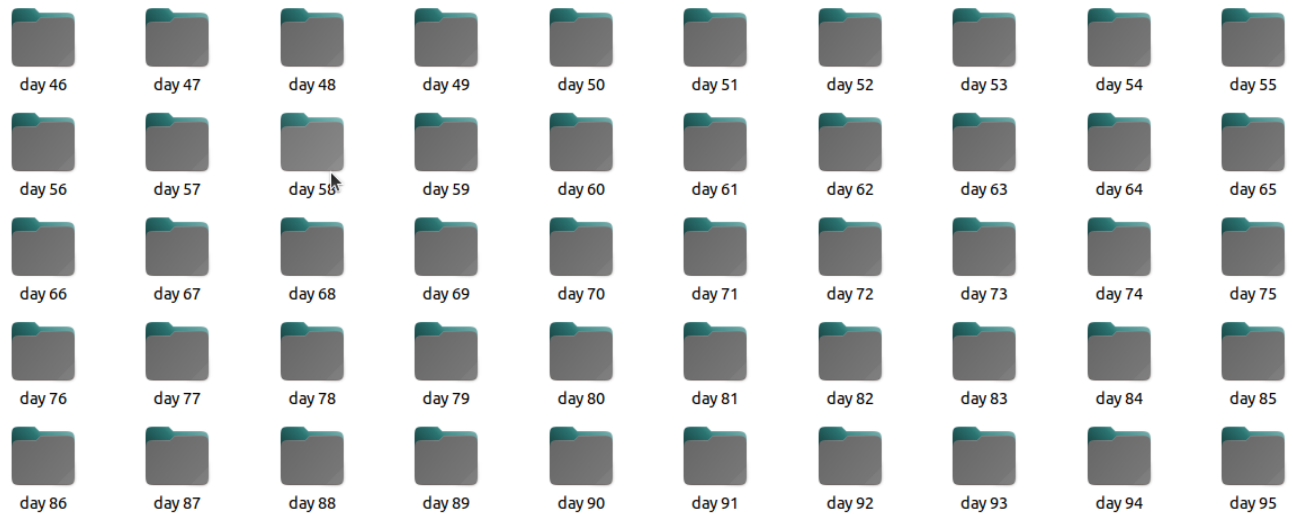
anayaazam

0 items

Task 6: Create Multiple Directories in a Loop

```
for i in range(1,100):  
    os.mkdir(f"day {i+1}")
```

Result:



Conclusions:

File handling operations (write, read, append), the `os.mkdir()` function, and the `__name__ == "__main__"` guard were practiced to understand how Python interacts with the operating system and modules.